WAEP Semester One Examination, 2015

Question/Answer booklet

MATHEMATICS METHODS UNIT 1

Section Two: Calculator-assumed

SOLUTIONS

Student number: In figures

In words ______________________________

Your name ______________________________

Time allowed for this section

Reading time before commencing work: ten minutes
Working time for section: one hundred minutes

Materials required/recommended for this section

To be provided by the supervisor
This Question/Answer booklet
Formula sheet (retained from Section One)

To be provided by the candidate
Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators approved for use in this examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of exam
Section One: Calculator-free	7	7	50	52	35
Section Two: Calculator-assumed	13	13	100	98	65
				Total	100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.

2. Write your answers in this Question/Answer booklet.

3. You must be careful to confine your response to the specific question asked and to follow any instructions that are specified to a particular question.

4. Additional working space pages at the end of this Question/Answer booklet are for planning or continuing an answer. If you use these pages, indicate at the original answer, the page number it is planned/continued on and write the question number being planned/continued on the additional working space page.

5. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.

6. It is recommended that you do not use pencil, except in diagrams.

7. The Formula sheet is not to be handed in with your Question/Answer booklet.

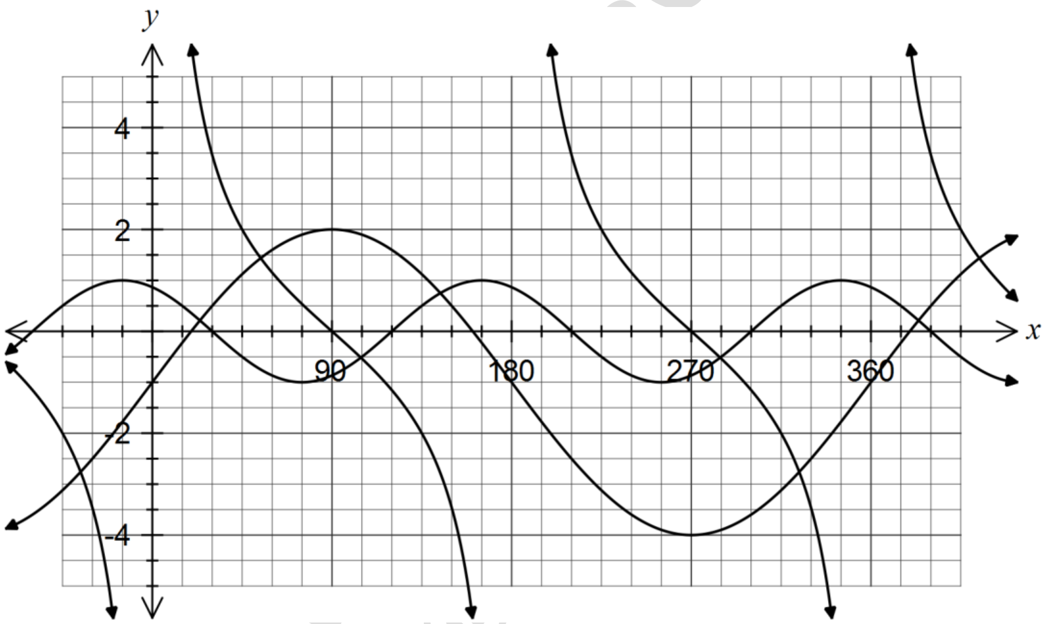
Section Two: Calculator-assumed **65% (98 Marks)**

This section has **thirteen (13)** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time for this section is 100 minutes.

Question 8 **(6 marks)**

The graphs of three functions, $f(x) = a\sin(x) + b$, $g(x) = c\tan\left(x + d\right)$ and $h(x) = \cos(px + q)$ are shown below.



Determine the values of the real constants a, b, c, d, p and q.

$a = 3$
$b = -1$

$c = -2$
$d = -90$

$p = 2$
$q = 30$

Question 9 **(8 marks)**

(a) The probability that a particular breed of bird, when in captivity, will develop an eye disorder is 3% and will suffer from feather mites is 7%. Assuming that these conditions are independent, what is the probability that one of these birds will develop

(i) neither condition whilst in captivity? (2 marks)

$$(1-0.03)\times(1-0.07)=0.97\times0.93$$
$$=0.9021$$

(ii) just one of these conditions whilst in captivity? (2 marks)

$$0.97\times0.07+0.93\times0.03=0.0679+0.0279$$
$$=0.0958$$

(b) A bag contains five red, three blue and four yellow balls. Two balls are selected at random from the bag without replacement. Determine the probability that

(i) the first is red and the second is blue. (1 mark)

$$\frac{5}{12}\times\frac{3}{11}=\frac{5}{44}$$

(ii) the balls chosen are blue and yellow, in any order. (1 mark)

$$\frac{4}{12}\times\frac{3}{11}+\frac{3}{12}\times\frac{4}{11}=\frac{2}{11}$$

(iii) neither ball is red. (2 marks)

Blue & Yellow: $\frac{2}{11}$

Blue & Blue: $\frac{3\times2}{12\times11}=\frac{1}{22}$

Yellow & Yellow: $\frac{4\times3}{12\times11}=\frac{1}{11}$

Total: $\frac{2}{11}+\frac{1}{22}+\frac{1}{11}=\frac{7}{22}$

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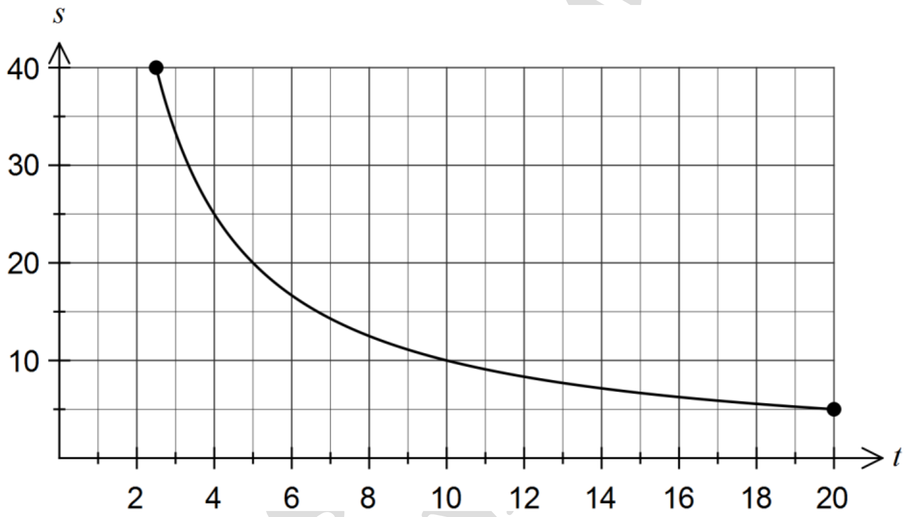
Question 10 **(8 marks)**

(a) The time, t seconds, that a motorcycle takes to travel a distance of 100 m is inversely proportional to its speed, s metres per second and is such that $s = \frac{100}{t}$.

(i) If the time doubles, what happens to the speed? (1 mark)

The speed halves.

(ii) Sketch the graph of speed against time for $2.5 \le t \le 20$. (2 marks)



(b) A function is given by $f(x) = \frac{3}{x+1} + 2$.

(i) State the natural domain and corresponding range of f. (2 marks)

Domain $\{x : x \in \mathbb{R}, x \ne -1\}$

Range $\{y : y \in \mathbb{R}, y \ne 2\}$

(ii) The graph of $f(x)$ is dilated vertically by a scale factor of 2. Determine the coordinates of the y-axis intercept. (2 marks)

Original y-intercept at (0, 5).

Hence new y-intercept at (0, 10).

(iii) The graph of $f(x)$ is translated 3 units to the right. Determine the equation of the translated function. (1 mark)

$$y = \frac{3}{(x-3)+1} + 2 = \frac{3}{x-2} + 2$$

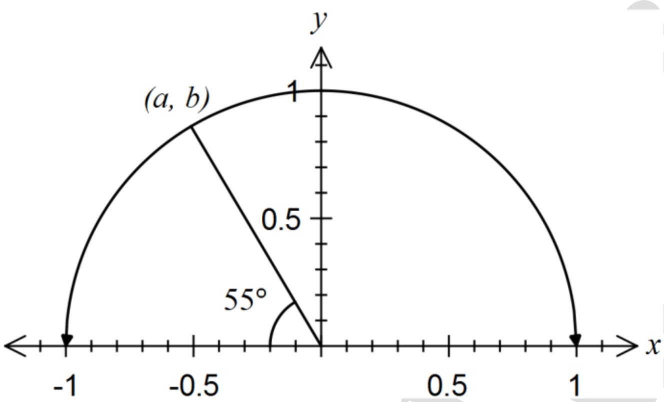
Question 11 **(6 marks)**

(a) A right-angled triangle ABC has a hypotenuse AC of length 8 cm and another side AB of length 6 cm. Determine, with justification, the exact value of $\tan\angle CAB$. (2 marks)

$$BC = \sqrt{8^2 - 6^2} = 2\sqrt{7}$$

$$\tan\angle CAB = \frac{2\sqrt{7}}{6} = \frac{\sqrt{7}}{3}$$

(b)



Part of a unit circle is shown above. Determine, in terms of a and b

(i) $\sin(235°)$. (1 mark)

$$\sin(235°) = -\sin(125°) = -b$$

(ii) $\tan(55°)$. (1 mark)

$$\tan(55°) = -\tan(125°) = -\frac{b}{a}$$

(c) A line is inclined at an angle of 120º to the positive x-axis and cuts the y-axis at (0, -2). Determine, with justification, whether the line passes through the point $(\sqrt{3}, -4)$. (2 marks)

$$m = \tan(150°) = -\sqrt{3}$$

$$y = -\sqrt{3}x - 2\Big|_{x=\sqrt{3}}$$
$$= -\sqrt{3}\sqrt{3} - 2$$
$$= -5$$

Hence line does NOT pass through $(\sqrt{3}, -4)$

Question 12 **(7 marks)**

(a) A box of chocolates contains ten different chocolates. Determine the number of different selections that can be made when

(i) two chocolates are chosen from the box. (1 mark)

$^{10}C_2 = 45$

(ii) one, two or three chocolates are chosen from the box. (2 marks)

One at a time: $^{10}C_1 = 10$

Two at a time: 45

Three at a time: $^{10}C_3 = 120$

Total: $10 + 45 + 120 = 175$

(b) 15 laptop computers are stored in a room, of which 11 are in working order but the rest have a fault. Determine

(i) the number of ways that four laptops could be chosen from the 15. (1 mark)

$^{15}C_4 = 1365$

(ii) the number of ways that three laptops could be chosen from the 11 laptops in working order. (1 mark)

$^{11}C_3 = 165$

(iii) the probability that when four laptops are selected at random from the room, three of them are in working order and one has a fault. (2 marks)

Choose 3 good and 1 faulty

$$\frac{^{11}C_3 \times {^4C_1}}{^{15}C_4} = \frac{165 \times 4}{1365}$$

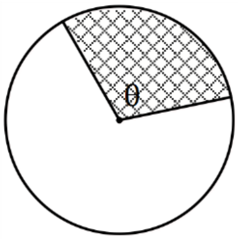
$$= \frac{44}{91}$$

$$(\approx 0.484)$$

Question 13 **(9 marks)**

(a) The shaded sector of the circle of radius 20 cm shown below has an area of 370 cm^2.

Determine the perimeter of the shaded sector. (2 marks)

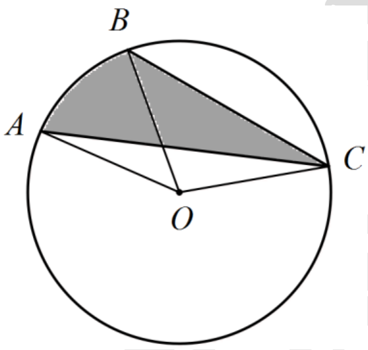


$$370 = \tfrac{1}{2}(20)^2\,\theta$$

$$\theta = 1.85$$

$$20 \times 1.85 + 20 + 20 = 77 \text{ cm}$$

(b) Determine the area of the shaded region ABC in the circle below with centre O and radius 24 cm, given that $\angle AOB = \frac{\pi}{5}$ and $\angle BOC = \frac{3\pi}{5}$. Give your answer to the nearest square centimetre. (3 marks)



Segment AC:

$$0.5(24)^2\left(\tfrac{4\pi}{5} - \sin\tfrac{4\pi}{5}\right) = 554.54$$

Segment BC:

$$0.5(24)^2\left(\tfrac{3\pi}{5} - \sin\tfrac{3\pi}{5}\right) = 268.96$$

Difference:

$$554.54 - 268.96 = 285.58$$

$$\approx 286 \text{ cm}^2$$

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(c) Given that triangle ABC has an area of 80 cm^2 and is such that $AB = 18$ cm and $AC = 16$ cm, one possible solution for the size of angle ACB is 84º, to the nearest degree.

Determine the other possible solution. (4 marks)

$80 = 0.5 \times 18 \times 16 \sin A \;\Rightarrow\; A = 33.7°$ or $180 - 33.7 = 146.3°$

$a^2 = 16^2 + 18^2 - 2(16)(18)\cos 33.7 \;\Rightarrow\; a = 10.05$

or

$a^2 = 16^2 + 18^2 - 2(16)(18)\cos 146.3 \;\Rightarrow\; a = 32.54$

$\dfrac{10.05}{\sin 33.7} = \dfrac{18}{\sin C} \;\Rightarrow\; C = 84°$

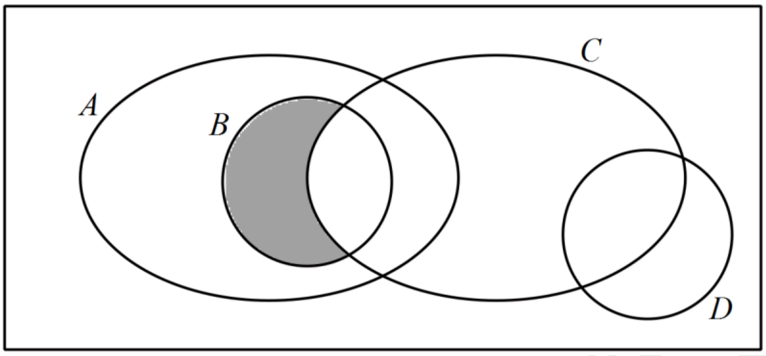
or

$\dfrac{32.54}{\sin 146.3} = \dfrac{18}{\sin C} \;\Rightarrow\; C = 18°$

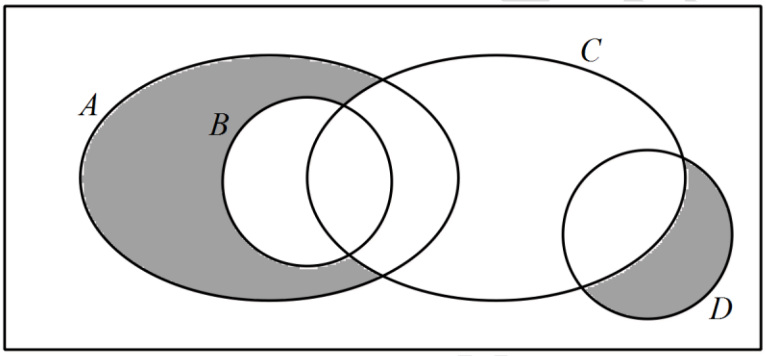
Question 14 **(9 marks)**

(a) Shade the regions on the Venn diagrams below to represent

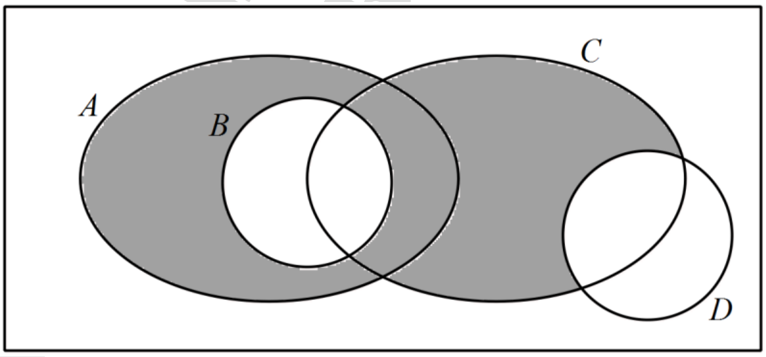
(i) $A \cap B \cap \overline{C}$. (1 mark)



(ii) $(A \cup D) \cap \left(\overline{C \cup B}\right)$. (2 marks)



(b) Use set notation to describe the shaded region in the Venn diagram below. (2 marks)



$(A \cup C) \cap \overline{B} \cap \overline{D}$

or

$(A \cup C) \cap \left(\overline{B \cup D}\right)$

(c) Two regular four-sided dice numbered from one to four are thrown. Event A occurs when the **sum** of the hidden faces is even and event B occurs when the **product** of the hidden faces is a multiple of three.

Explain, with justification, whether events A and B are independent, mutually exclusive or neither. (4 marks)

Event A

A	1	2	3	4
1	2	3	4	5
2	3	4	5	6
3	4	5	6	7
4	5	6	7	8

Event B

B	1	2	3	4
1	1	2	3	4
2	2	4	6	8
3	3	6	9	12
4	4	8	12	16

$P(A)=\frac{1}{2}$

$P(B)=\frac{7}{16}$

$P(A\cap B)=\frac{3}{16}$ $\Rightarrow$ Not mutually exclusive

(Or observes that (3,1) $\in A, \in B$, etc)

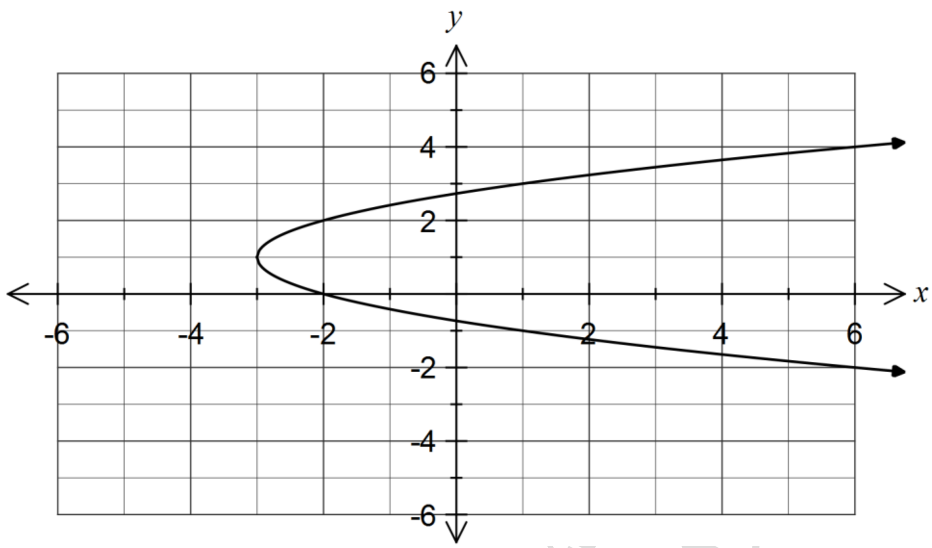
$P(A)\times P(B)=\frac{7}{32}$

$P(A)\times P(B)\neq P(A\cap B)$ $\Rightarrow$ Not independent

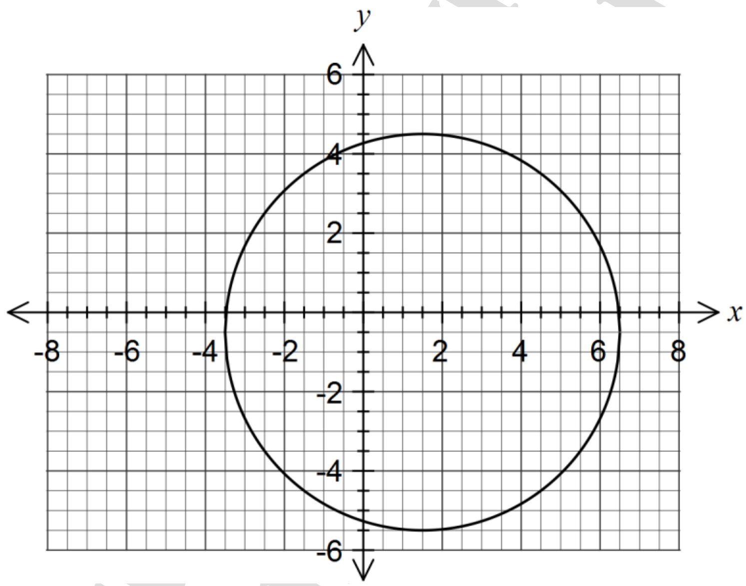
Hence neither

Question 15 **(6 marks)**

(a) Sketch the graph of $x+3=(y-1)^2$. (2 marks)



(b) Determine the equation of the circle shown below. (2 marks)



$$(x-1.5)^2+(y+0.5)^2=25$$

(c) Determine the radius of the circle with equation $x^2-4x+y^2+6y+8=0$. (2 marks)

$$x^2-4x+y^2+6y+8=0$$
$$(x-2)^2-4+(y+3)^2-9=-8$$
$$(x-2)^2+(y+3)^2=5 \Rightarrow r=\sqrt{5}$$

Question 16 **(6 marks)**

(a) A and B are both obtuse angles such that $\cos A = -\frac{4}{5}$ and $\tan B = -\frac{1}{2}$. Determine an exact value for $\cos(A-B)$. (3 marks)

$$\sin A = \frac{3}{5} \qquad \sin B = \frac{1}{\sqrt{5}} \qquad \cos B = -\frac{2}{\sqrt{5}}$$

$$\begin{aligned}\cos(A-B) &= \cos A \cos B + \sin A \sin B \\ &= -\frac{4}{5} \times -\frac{2}{\sqrt{5}} + \frac{3}{5} \times \frac{1}{\sqrt{5}} \\ &= \frac{11\sqrt{5}}{25}\end{aligned}$$

(b) State the exact values of $\sin 45°$ and $\cos 120°$, and hence show that the exact value of $\sin(165°)$ is $\frac{\sqrt{2}\left(\sqrt{3}-1\right)}{4}$. (3 marks)

$$\sin 45 = \frac{1}{\sqrt{2}} \qquad \cos 120 = -\frac{1}{2}$$

$$\begin{aligned}\sin(165) &= \sin(45+120) \\ &= \sin 45 \cos 120 + \cos 45 \sin 120 \\ &= \frac{1}{\sqrt{2}} \times -\frac{1}{2} + \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} \\ &= \frac{\sqrt{3}-1}{2\sqrt{2}} \\ &= \frac{\sqrt{2}\left(\sqrt{3}-1\right)}{4}\end{aligned}$$

Question 17 **(9 marks)**

The table below summarises the age group and main type of injury (head or other) of cyclists who required hospitalisation following an accident over the periods one year before and one year after the implementation of a bike-sharing scheme (BSS) in a city.

	Before BSS			After BSS		
Age	Head	Other	Total	Head	Other	Total
<15	72	123	195	54	55	109
15-24	60	58	118	42	30	72
25-49	97	137	234	99	112	211
≥ 50	71	102	173	58	63	121
Total	300	420	720	253	260	513

(a) What percentage of cyclists in the above data sustained a head injury over the two year period? (1 mark)

$$\boxed{\frac{300+253}{720+513}\times 100 = 44.85\%}$$

(b) If one cyclist was selected at random from the data, determine the probability that

(i) they sustained a head injury after the implementation of the bike-sharing scheme and were aged between 25 and 49. (1 mark)

$$\boxed{\frac{99}{1233}\ (\approx 0.0803)}$$

(ii) they sustained an 'other' injury before the implementation of the bike-sharing scheme or were aged over 24. (2 marks)

$$\boxed{\frac{420+97+71+211+121}{1233} = \frac{920}{1233}\ (\approx 0.7461)}$$

(iii) their accident occurred after the implementation of the bike-sharing scheme given that they were aged under 50. (2 marks)

$$\boxed{\frac{513-121}{1233-173-121} = \frac{392}{939}\ (\approx 0.4175)}$$

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(c) Does the above data suggest that the chance of a cyclist sustaining a head injury in an accident is independent of age after the introduction of the bike-sharing scheme?

Support and justify your answer with relevant calculations. (3 marks)

P(HI After BSS) = $\frac{253}{513} \approx 0.493$

P(HI After BSS | Aged 15-24) = $\frac{42}{72} \approx 0.583$

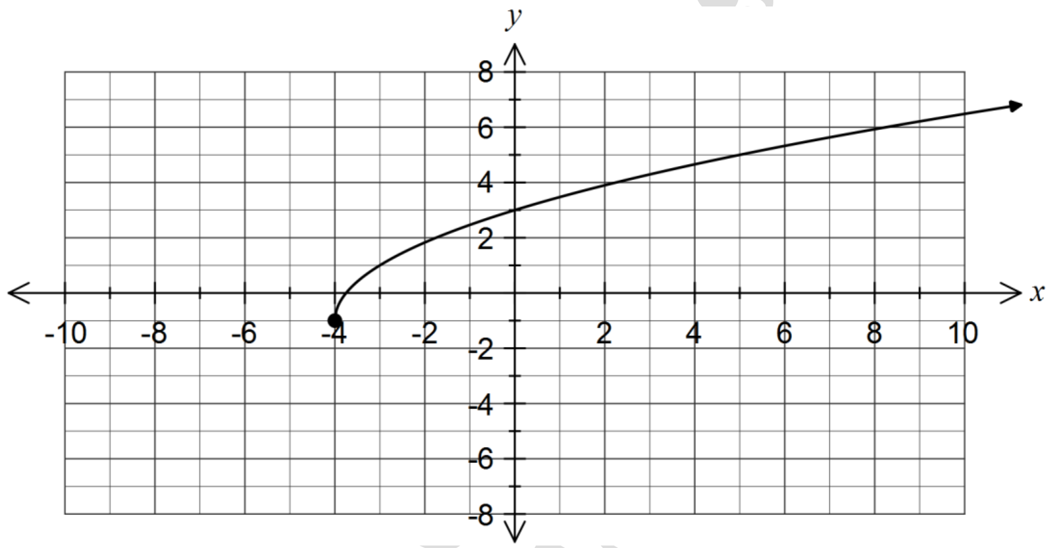
No, not independent.

Although the conditional probabilities for the other age groups are all close to 0.493, for the 15-24 age group, the probability is significantly higher (0.583) and so we conclude that sustaining a head injury is not independent of age.

Question 18 **(7 marks)**

(a) The function f is defined by $f(x) = 2\sqrt{x+4} - 1$.

(i) Sketch the graph of $y = f(x)$. (2 marks)



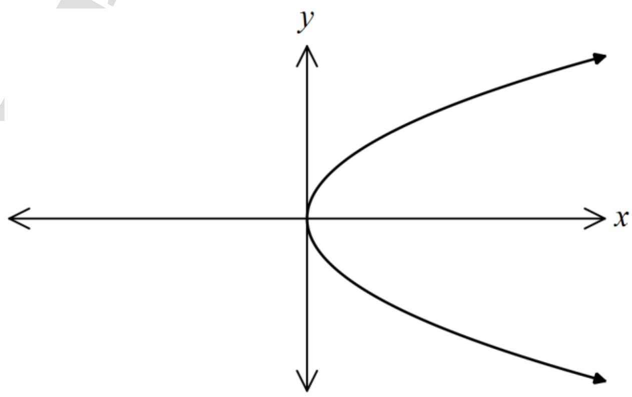
(ii) Describe how to transform the above graph to obtain the graph of $y = f(x+1)$. (1 mark)

Translate graph one unit to the left.

(iii) Describe how to transform the graph of $y = f(x)$ to obtain the graph of $y = 2 + f(-x)$. (2 marks)

Reflect graph in the y-axis and translate upwards by two units.
(Order not important in this case)

(b) On the axes below, sketch the graph of an example of a relation that is clearly **not** a function and explain why it is not a function. (2 marks)



A vertical line drawn through the graph for $x > 0$ cuts the graph in more than one place – hence not a function as a one to many relationship exists.

Question 19 **(9 marks)**

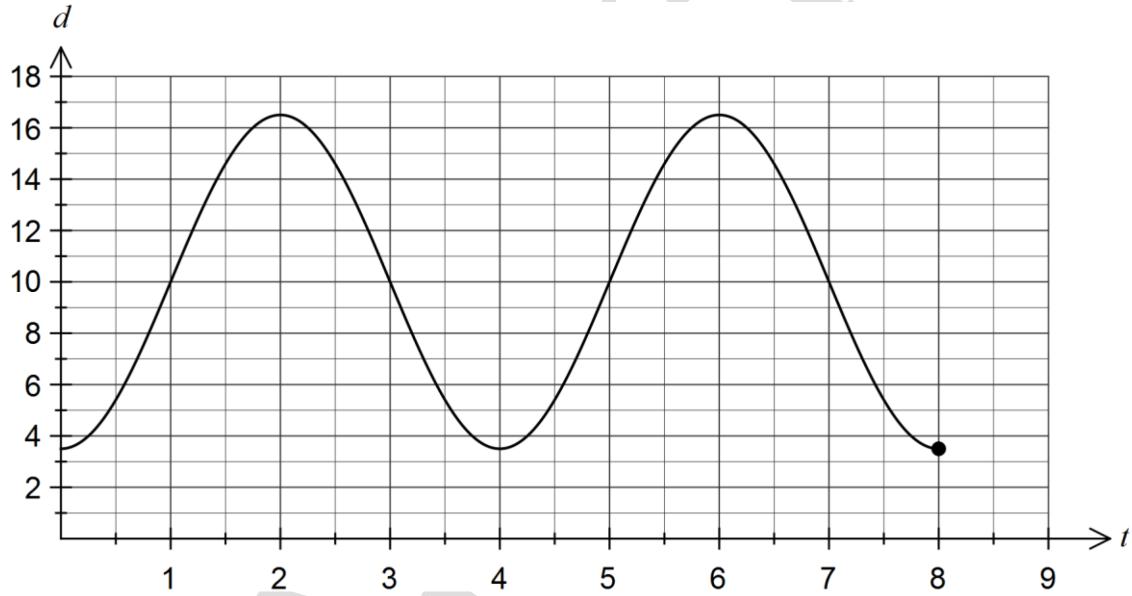
A small object is moving in a straight line towards and away from a fixed point O.

The horizontal distance, d m, of the object from O at time t seconds is given by the function $d(t) = 10 - 6.5\cos\left(\frac{\pi t}{2}\right)$.

(a) Determine the distance of the object from O when $t = 3$. (1 mark)

10 m

(b) Draw the graph of $d(t)$ on the axes below for $0 \le t \le 8$. (3 marks)



(c) State the amplitude and period of the graph of $d(t)$. (2 marks)

Amplitude is 6.5 m.

Period is 4 seconds.

(d) Determine the time, to the nearest tenth of a second, when the object was first at a distance of 16 metres from O. (1 mark)

$t = 1.7487$

≈ 1.7 to nearest tenth of second

(e) Determine the distance travelled by the object between $t = 2$ and $t = 22$. (2 marks)

$\frac{22-2}{4} = 5$ complete cycles

$5 \times 4 \times 6.5 = 130$ metres

Question 20 **(8 marks)**

For two events, A and B, $P(A) = \frac{3}{5}$ and $P(A \cup B) = \frac{5}{6}$.

(a) Determine the minimum possible value of $P(B)$. (2 marks)

Minimum if A and B are disjoint.

$$P(B) = \frac{5}{6} - \frac{3}{5}$$
$$= \frac{25-18}{30} = \frac{7}{30}$$

(b) Determine the maximum possible value of $P(A|B)$. (3 marks)

Maximum when A is a subset of B.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$
$$= \frac{P(A)}{P(A \cup B)} \text{ when } A \text{ is a subset of } B$$
$$= \frac{3}{5} \div \frac{5}{6} = \frac{18}{25}$$

(c) Determine the value of $P(A \cap B)$ if A and B are independent. (3 marks)

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

But $P(A \cap B) = P(A)P(B)$ when independent

$$\frac{5}{6} = \frac{3}{5} + P(B) - \frac{3}{5} \times P(B)$$
$$P(B) = \frac{7}{12}$$

$$P(A \cap B) = \frac{3}{5} \times \frac{7}{12} = \frac{7}{20}$$

End of questions

Additional working space

Question number: _________